Paths:

We have 2 different types of paths:
1)Absolute Path : The path which is reference to the root(/) directory

#mkdir –p /sap/data/db

#cd /sap/data/db

#pwd
O/P
/sap/data/db

2) Relative Path : The path which is reference to the home or current working directory other than root.

#mkdir -p /sap/test/objects

#pwd
O/P
/sap/data/db

#cd ../../test/objects

#pwd
O/p
/sap/test/objects

(Displays inode number of specific file)
#ls -i <FN>
Abc1

O/P
6568 abc1

#ls -R /sun6
(Displays FS available Inodes Information)

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#df -i

UMASK (User Mask value) :

The purpose of the user mask valus is to provide security for both files & Directories.

NOTE: The default UMASK value is 022

Perm values :
read --> 4
write -->2
execute --->1

RWX --->7

When we create file Perm given by OS ---> 666 (rw- rw- rw-)
When we create dir perm given by OS ----> 777 (rwx rwx rwx)

How umask provides Security :

For Files : 666
(-) 022
==========
Final File Perm of File - 644 (rw- r-- r--)
==========

For Directories : 777 (rwx rwx rwx)
(-) 022
==========
Final File Perm of Directory - 755 (rwx r-x r-x)
==========

CHMOD (CHange Mode) :

The purpose of CHMOD command is to change the perm of files / directories.
NOTE: Chmod allows us to change the common permissions.

We can use CHMOD comman in 2 modes :

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1)Numeric Method
2) Symbolic mode

Numeric Method :

Perm values :
read --> 4
write -->2
execute --->1

RWX --> 7

#touch file15

#ls -l file15

O/P
-rw- r-- r-- (6 4 4)

Target is to change the perm of file15 from -rw- r-- r-- (6 4 4) to rwx rw- r-x (7 6 5)

#chmod 765 file15

#ls -l file15
O/P
-rwx rw- r-x

Absolute Method:

We are using alphabets & Special Characters

+ --> Adding PERM

- ---> removing the PERM

u --> Owner Perm
g -->Group Perm
o --> Others Perm

#chmod u+x,g+w,o+x file18

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Inode Number:

It is the unique number assigned to every file. For every file it has file name and inode number.File name is for user reference and inode number is for kernel reference.
NOTE: Inode number is unique for every file.

NOTE : Inode number contains every information about the files like :

Owner of the file
Group owner of the file
Perm of the file
Number of links
Size of the file

```
#ls     -i      <FN>
                 Abc1
```

O/P
6568 abc1

LINKS:

Links are pointers
Connection between two strings

2 types of Links

1)Hard Links
2)Soft Links

Hard Links:
The two files having hard links will have the same inode number. If we update the data to the source file the same data we can access from the destination file & vice versa.
If we delete the source file we can access the data from the destination file with the same inode number. Hard links works as duplicate copies.

Limitations:
1)We cannot identify the files having hard links
2)hard links are applicable for only files
3)Hard links need to be configured with in the same filesystem.

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#ln <SF> <DF>
Cap11 cap12

Soft Links :

The two files having soft link will be having different inode number. If we update the data to the source file the same data we can access from the destination file & vice versa.

If we delete the source file we cannot access the data from the destination file.Soft links acts as short cuts.

Advantages:

1)We can identify the files having soft links by seeing its perm.
2)Soft links are applicable for both files & directories.
3)Soft links can span across multiple filesystems.

#ln -s <SF> <DF>
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